PREDICTIVE MOVEMENT MODELING USING HMMS AND DBNS

Akash Rana

CSDS 491 - Project Report

OBJECTIVE

01

Forecast user's next location and activity in smart homes.

02

Use HMM to model sequential dependencies.

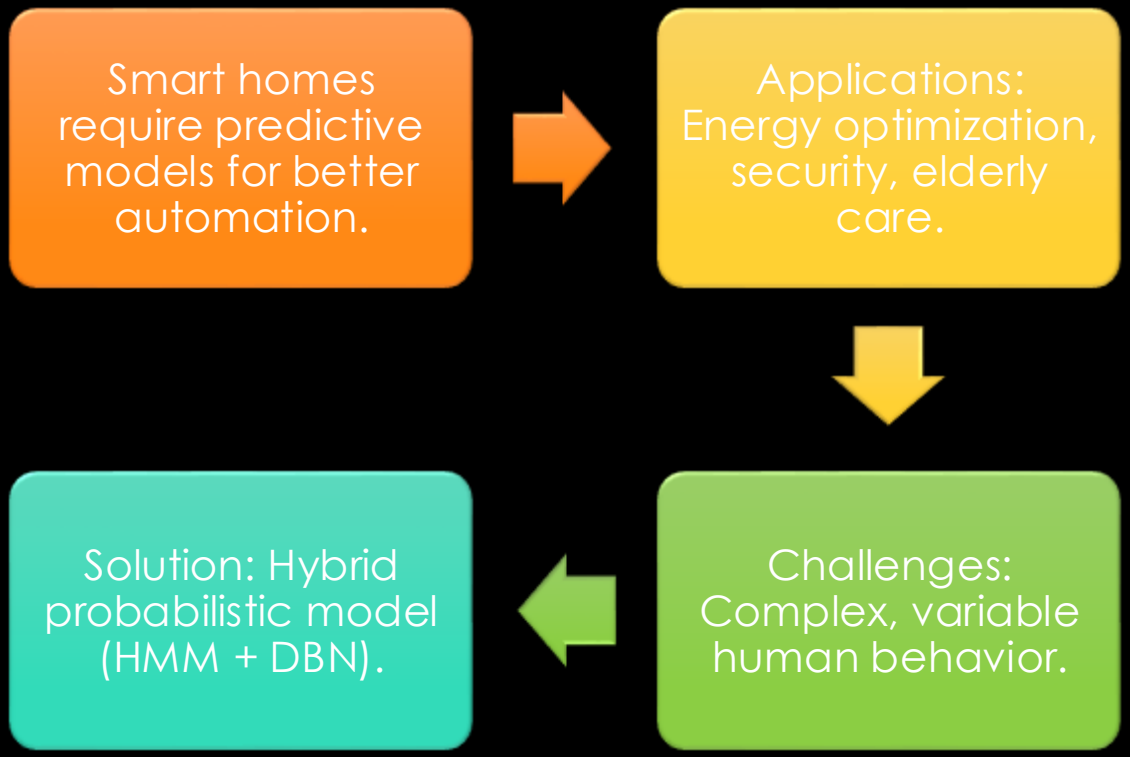
03

Use DBN for richer temporal dynamics and contextual variables.

04

Support smart home automation, energy management, and security.

INTRODUCTION & MOTIVATION



RELATED WORK

HMMs: Model temporal dependencies in user behavior.

DBNs: Extend HMMs with contextual info (time, environment).

Deep learning models (LSTM, GRU): Data-intensive, not real-time.

Selected HMM and DBN for interpretability and efficiency.

DATASET OVERVIEW

Sensor activations with timestamps.

Attributes: Time, Sensor ID, Room Label, Temperature, Time of Day.

Context-rich sequential data supports probabilistic models.

PREPROCESSING PIPELINE

Map sensor IDs to room labels.

Extract time of day and temperature features.

Chronological sequencing and cleaning.

Remove noise and ensure temporal continuity.

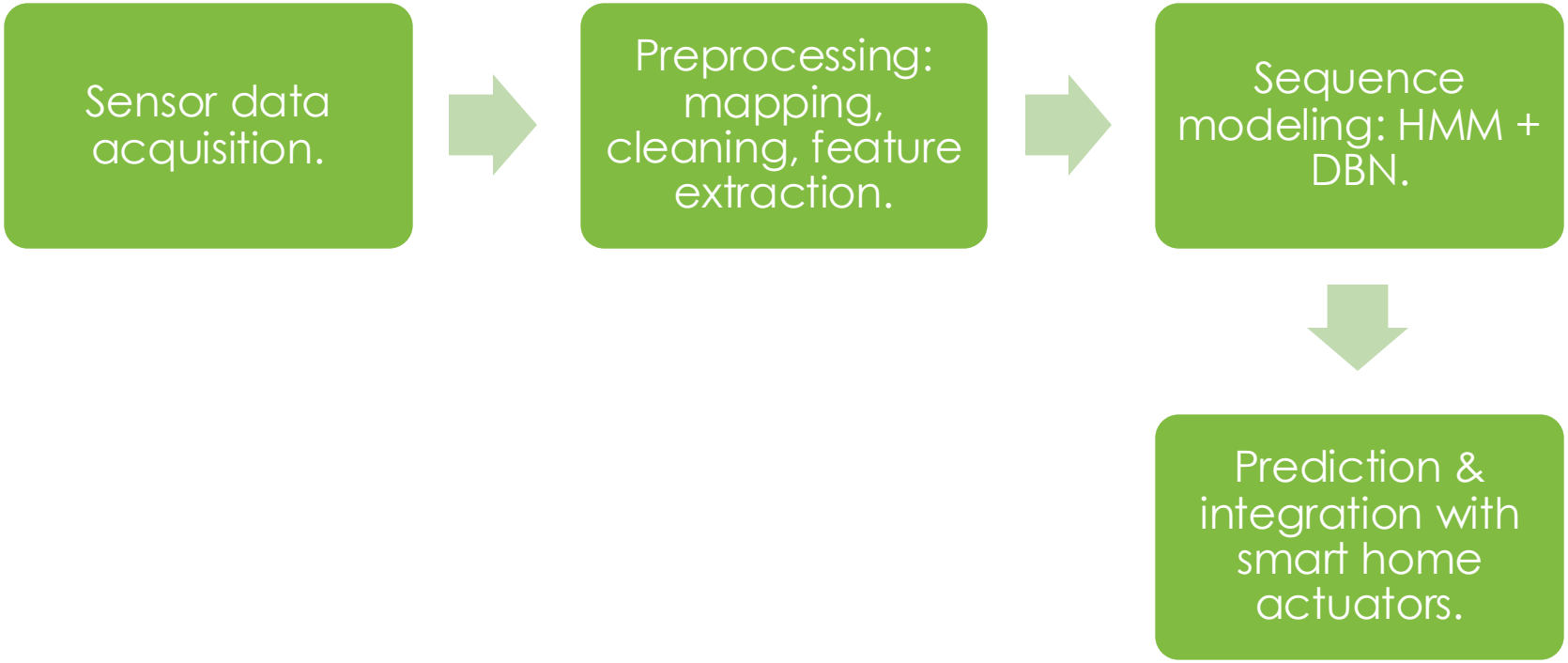
THEORETICAL FOUNDATIONS: HMM

- Generative probabilistic model with hidden states.
- States: User's intent (active, idle, resting).
- Observations: Location + Activity.
- Algorithms: Forward-Backward, Baum-Welch, Viterbi.

THEORETICAL FOUNDATIONS: DBN

- Extension of HMM with contextual variables.
- Models joint probability of activity and location.
- Training: EM Algorithm.
- Inference: Particle Filtering for real-time updates.

SYSTEM ARCHITECTURE



WORKFLOW

Data Collection →Preprocessing → Model Training → Inference & Prediction.

HMM captures temporal sequences.

DBN refines with context-aware predictions.

Outputs predictions for smart home automation.

EVALUATION STRATEGY

Training/testing sequences with temporal integrity.

Metrics: State Prediction Accuracy, Observation Accuracy, Sequence Accuracy.

Contextual Consistency for time and temperature variables.

RESULTS SUMMARY

- HMM predicts frequent patterns well.
- DBN improves context-aware predictions.
- Sequence accuracy high across longer sequences.
- Robust performance during routine transitions.

CHALLENGES AND SOLUTIONS

Data Sparsity: Laplace smoothing, synthetic data.

Overfitting: Regularization (AIC/BIC).

Temporal Alignment: Discretization of time.

Noisy Data: Cleaning routines.

FUTURE WORK

- Expand to larger, real-world datasets.
- Enhance DBN with granular context inputs.
- Real-time integration with smart home systems.
- Scalability and multi-user modeling.

CONCLUSION

- Hybrid HMM-DBN effectively predicts user behavior.
- Contextual features enhance accuracy.
- Supports real-time smart home automation.
- Foundation for future intelligent environments.

REFERENCES

- CASAS Smart Home Project - WSU
- L. R. Rabiner, 'HMM Tutorial', IEEE 1989
- K. P. Murphy, 'Dynamic Bayesian Networks', UC Berkeley
- Hmm learn, Pyro libraries
- D. J. Cook et al., Smart Home Activity Recognition
- G. Lazarou et al., Smart Home Energy Management